Yield & atom economy

Predict & measure

1 Theoretical maximum

mass | moles | ratio

Convert → compare → convert back

Use mass → moles → equation ratio → product mass. Check limiting reactant first.

Example: $CaCO_3 \rightarrow CaO + CO_2$. 25.0 g $CaCO_3$ / 100 = 0.250 mol → 0.250 × 56 = **14.0 g CaO.**

2 Percentage yield

Yield = actual / theoretical × 100%. If actual CaO is 11.2 g: 11.2/14.0 × 100 = **80.0%.**

Actual = theoretical × yield/100. Theoretical = actual / (yield/100).

Account for atoms

3 Why yield is lower

Reversible reaction may not finish; side reactions consume reactants; product is lost in transfer/separation.

Atoms are still conserved. Apparent yield above 100% suggests wet/impure product or measurement error.

4 Atom economy

Desired-product coefficient × M_r divided by sum of all reactant coefficient × M_r; then × 100%.

$CaCO_3 \rightarrow CaO + CO_2$: for desired CaO, 56/100 × 100 = **56%.**

Other 44% becomes CO_2, even if every atom reacts correctly.

Choose a pathway

5 Keep the ideas separate

Yield: proportion of the expected product actually collected. Atom economy: proportion of starting mass directed into desired product.

A process can have high yield but low atom economy. Addition reactions with one product have 100% atom economy.

6 Evidence → decision

Higher atom economy reduces waste and resource/disposal costs. But compare yield, rate, energy, equilibrium and useful by-products too.

Support a choice using the supplied data and acknowledge a trade-off. Repeats/uncertainty help judge measured yield.